UTHM EA

SECP3744-01

UTHM Enterprise Architecture

Prepared by SysDataverse

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AS-IS Architecture

Architecture Principles, Vision, and Requirements

Architecture Vision

Preliminary
- Architecture
- Principles

Continuously upgrade technology and phase out outdated legacy systems and manual methods so they are no longer required.

Achieve full "Digital Transformation" and improve the overall user experience by removing all manual processes and moving toward complete end-to-end digitalization.

Stakeholders
- Students
- Faculty
- Admin
- Ministry + Industry Partners
- Accreditation Bodies

Architecture Requirements

- University KPI to drive digitalization efforts.
- User complaints about legacy client-server systems must be addressed.
- If security threats occur and no immediate action is taken, the ICT team must shut down the system for safety.

Business Architecture

Motivation

Drivers
- Need to align with national digitalization goals

Goals
- Build digital first culture
- Improve user experience
- Update to the latest technology

Objectives
- More than 82% digital services by 2030

Measures
- % of successfully digitalized services

Organization

Organization Structure
- University Leadership (Board, Senate, Vice Chancellor)
- Academic Divisions (Faculties, Schools, Academic Departments)
- Administrative Divisions (Student Affairs, Finance, HR, Registrar)
- Research & Innovation Units
- Campus Operations (ICT Center, Facilities, Library)

Actors & Roles
- Student
- Lecturers
- Researchers
- Admin
- Staff
- Student from IPTA/IPTS
- Public & Private Sector
- Non-Governmental Organizations
- General Public

Behavior

Business Services
- Academic services
- Research services
- Multimedia/digital content services
- Assessment services
- Staff training services

Processes
- Enrollment & registration
- Proposal/approval workflows
- Content creation & capture
- Exam creation & grading
- Training scheduling & tracking

Functions
- Academic mgmt
- Research mgmt
- Multimedia mgmt
- Assessment mgmt
- Training & HRD mgmt

Information System Architecture

Data

Data Entities

Students
Staff
Courses
Programmes
Enrollments
Assessments
Exam Grades
Learning Content
Payment Record
Counseling Record
Staff Training Record

Data Usage

SMAP → Student & Curriculum
AUTHOR → Learning Content
MOOC/e-Kursus → Learning Content
CMS → Student & Counseling Record
SAS → Assessment & Grades
SPPLN/IMPAK → Staff
UTHM Identity → User Data
Student Email Service → Student Data

Application

Application Services

SMAP (SIS)
AUTHOR (LMS)
AUTHOR ODL (LMS)
MOOC UTHM
e-Kursus
CMS
DIGITAL UTHM
SPPLN
IMPAK
SAS
SPPU
UTHM Identity (SSO & 2FA)
Student Email Service

Logical Components

Identity & Access Management
Student Information System (SIS)
Learning Management System (LMS)
Assessment & Examination Management
Student Support Services
Staff Management
Campus Operations Management

Technology Architecture

Technology Domain
- Application platform
- User access
- Data management
- Physical infrastructure
- Network performance
- Authentication
- Security controls
- Software development

Architectural Realization

Opportunitites, Solution and Migration Planning

Opportunities
- System integration
- Eliminate manual process
- Improve system accessibility
- Data consolidation

Solution
- Application platform
- Core systems
- Integration method
- Cyber security

Migration Planning
- Take 3 years to fully migrate
- ~72% of migration is currently complete
- Follow approval process

Implementation Governance

Standards
- ISO 27001 information security
- Database naming conventions
- API integration standards

Guidelines
- Data privacy guidelines
- Password policy
- Backup and retention guidelines

Specifications
- Database size & growth plan
- Bandwidth for peak usage
- Dual-firewall architecture

Architecture Principles, Vision and Requirements

Principles

- Replace outdated systems
- Reduce manual work

Vision

- Achieve full digital transformation
- Improve user experience
- Ensure all services are connected

Requirements

- Collect and consider users' feedback
- Maintain high system security
- ICT team must respond immediately to any security issues

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AS-IS Problems

Current State

Manual & Paper Based Process

- PKU relies heavily on manual documentation
- Slow patient registration
- Difficulty in tracking medical histories

Analytics & Business Intelligence Issues

- UTHM has massive amounts of data
- Data trapped in separate systems
- No Real-Time view for university management

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To-Be Business Architecture

STRATEGY

- Driver

External pressure for 'Always Online' systems. Need for real-time analytics to replace manual reporting.

- Goals

Establish a Hybrid Architecture: Combining On-Premise security with Cloud scalability

- Objective

100% Paperless & Cashless. Zero physical forms for administrative tasks.

- Measures

99.99% availability for LMS

ORGANIZATION

- New hybrid unit

Hybrid Infrastructure & Integration Unit to bridge legacy with cloud.

- New roles

Integration Specialists & Data Center Ops Team.

SERVICES

- Personalized Learning

AI driven content adaptive to student needs.

- Smart Campus

IoT-enabled Parking & Library access.

PROCESS & FUNCTIONS

- Automated Enrollment

Instant regulation without human approval delays.

- AI-Protocol Exams

Autopiloted online assessment ensuring integrity.

- Hybrid Operation Mangement

Single-pane view of IT health (Cloud + On-Premise).

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To-Be Application Architecture

SERVICE

- Learning Platforms
- Administrative Platforms
- Enterprise Services

LOGICAL COMPONENT

- **Identity & Access Management:** Authentication and user rights.
- **Student Information System (SIS):** This is the database of student lifecycle.
- **Learning Management (LMS):** The teaching, learning, and collaboration interface.
- **Assessment and examination:** Intensity course modules: grading and exam integrity.
- **Campus Operations:** Systems: Research and day-to-day logistical systems.
- **Health Management**: A safe university medical service system.

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To-Be Data Architecture

DATA ENTITIES

- Essential Academic Branches: Students, personnel, courses, and programs.
- Academic history: classifications, examinations, and grades.
- Operational Assets: Learning material and payment history.
- **Sensitive Records**: Special training and medical history records.

DATA USAGE

- LMS Integration
- Administrative Management
- Security & Identity
- **Strategic Intelligence:** Business intelligence (BI) tools combine data to provide a management dashboard.
- **Governance:** The medical data access is highly divided and limited to the privacy policies.

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To-Be Technology Architecture

Centralized User Access
- One system manages all user accounts
- Standard access rules, easy onboarding/offboarding

Managed Data Services
- Central data storage
- Regular backups, consistent data, easy reporting

Hybrid Infrastructure
- Mix of on-premise + cloud
- Sensitive data on-premise, scalable apps on cloud

Optimized Network
- Faster internet & strong WiFi
- Network segmentation for safety
- Stable online learning

Fully Integrated SSO
- One login for all systems (SIS, LMS, HR, Library, Finance, PKU)
- Central identity management
- Ready for MFA

Enhanced Security
- Firewalls, monitoring, encryption
- PDPA compliance
- Zero Trust approach
- Regular security audits

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Gap Analysis and Roadmap

GAP ANALYSIS

CURRENT:

- Mostly digital process, but 20-30% remain manual.
- Siloed legacy applications causing performance issues.
- Infrastructure mostly on-premise lead to bottlenecks during peak usage.

TO-BE:

- Fully digital and paperless.
- Integrated systems with Master Data Management.
- Hybrid cloud infrastructure to handle high-traffic applications

ROADMAP

- **Short-term:** rationalize systems & consolidate processes into digital modules.
- **Medium-term:** deploy a hybrid cloud, strengthen connectivity & enhane security.
- **Long-term:** implement smart campus IoT services and automate workflows.

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Conclusion

Values

- Efficient & Paperless Operations
- Integrated & Secure Systems
- Better Decisions & User Experience

Manual processes, siloed systems, and limited analytics visibility.

The proposed TO-BE Enterprise Architecture introduces a **hybrid cloud model** that balances security, scalability, and performance.

Q&A Session

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